Zewail City of Science, Technology and Innovation
University of Science and Technology
Communications and Information Engineering
CIE 599
Evaluation Rubric 2025/2026

Section-by-section evaluation (detailed assessment of all elements of the report) – Per Team – Examining Committee – 49%

	Element	Outstanding (4)	Meets Expectations (3)	Needs Improvement (2)	Inadequate (1)
Abstract and Introduction (5%)					
1	The introduction is written in a clear and readable style that is appropriate to non-technical audience.				
2	The introduction presents the motivation, key challenges, project scope, and proposed methodology in a clear and organized manner.				
3	The introduction considers the impact of the problem in global, economic, environmental, and societal contexts (all the four contexts must be considered - the project may focus on a subset of these elements).				
4	The abstract is written in a concise and readable style that is appropriate to non-technical audience.				
Literature Review (13%)					
5	The report demonstrates the ability of students to recognize the recent and ongoing research, technical advancements, and changes in technology based on the appropriate sources of information.				
6	The report demonstrates the ability of students to recognize the Code of Ethics for the discipline.				
7	The report provides a review of the related codes, regulations, and laws.				
8	The report provides a review of the standards that can be used in the design process.				

9	The report provides a recent and relevant market survey.				
Proposed Design (15%)					
10	Students are able to write a clear and concise problem statement.				
11	Students are able to identify requirements, constraints and risks.				
12	Students are able to identify standards and regulations affecting the design.				
13	The proposed design considers public health, safety, welfare, as well as global, cultural, social, environmental, and economic factors.				
14	The report demonstrates the ability of students to propose multiple solutions.				
15	The report demonstrates the ability of students to evaluate the different solutions against requirements.				
16	The report demonstrates the ability of students to use a decision analysis technique to select the proper design.				
17	The report demonstrates the ability of students to make informed decisions that adhere to recognized codes and laws.				
18	The report demonstrates the ability of students to make informed decisions that consider any major concerns in global, economic, environmental, and societal contexts.				
Implementation (15%) - advisor					
19	The report demonstrates the ability of students to develop an initial implementation of the selected design.				
20	The report demonstrates the ability of students to use iterative process to improve the implementation.				
21	The report demonstrates the ability of students to determine data that is appropriate to collect or use, and select appropriate sources, equipment, protocols, etc., for measuring the appropriate variables.				

22	The report demonstrates the ability of students to conduct an experiment or simulation in compliance with appropriate test standards and operate instrumentation and/or software packages appropriately.				
23	The report demonstrates the ability of students to use appropriate computer-based tools, statistics, and other resources effectively to analyze, validate, and interpret experiment results.				
24	The students are able to incorporate appropriate Engineering standards in their implementation.				
25	The report demonstrates the ability of students to compare results to theoretical predictions.				
26	The report Identifies the major impact of the project in global, economic, environmental, and societal contexts.				
27	The report demonstrates the ability of the students to select the performance metrics and to create a comprehensive testing report using the appropriate test cases.				
Conclusion (1%)					
28	The report demonstrates the ability of students to draw conclusions using engineering judgment.				

Overall assessment of the project idea – Per Team – Examining Committee – 8%

	Element	Outstanding (4)	Meets Expectations (3)	Needs Improvement (2)	Inadequate (1)
29	The project proposes and provides solutions addressing problems that are of practical significance or with national impact.				
30	The project proposes and provides an innovative solution, leading to a possible commercial product, a patent, or to a research publication.				
31	The project was able to seek and gain support from industry/organization (technical, financial, …).				
32	The project was able to propose a multi-disciplinary solution.				
33	The project demonstrates the ability of the student to gain and apply new knowledge.				

Assessment of the project demo – Per Team – Examining Committee – 15%

Project Demo (15%) – committee					
	Element	Outstanding (4)	Meets Expectations (3)	Needs Improvement (2)	Inadequate (1)
34	The project demo successfully demonstrates the functional requirements identified in the report. (weight x 2)				
35	The project demo successfully demonstrates the non-functional requirements identified in the report. (weight x1)				
36	The project demo demonstrates the ability of the students to follow the clean code guidelines for software products. For hardware products, the students were able to use proper schematics, properly sized and sturdily mounted components, appropriately enclosed prototypes with no bare wires or loose plugs, and appropriately labeled subsystems. Batteries are suitable for the application. Safety and emergency keys, safety circuit breakers and fuses, brakes, etc., fail safe scenarios were considered. (weight x0.5)				
37	The project demo demonstrates the ability of the students to develop a user-friendly product. Also properly select the user interface, size and type, and location/position of the user interface, use of prober key board labels, color, size, and resolution of screens, screen layout, navigation menu with clear instructions. The software is written with comments, with proper safe code and written in modular parts with structure and documentation for GitHub code sharing. (weight x0.5)				

Assessment of the technical writing skills – Per Team – Examining Committee – 3%

	Element	Outstanding (4)	Meets Expectations (3)	Needs Improvement (2)	Inadequate (1)
38	The report communicates the ideas in an organized flow in clear and readable style that is appropriate to the technical audience.				
39	The report communicates the ideas in mostly (grammatical and spelling) error-free language; and use effective, clear visual aids.				
40	The report provides an original, unique text; and appropriately cite the used sources following a proper format.				

Presentation Skills – Per Team – Examining Committee – 10%

	Element	Outstanding (4)	Meets Expectations (3)	Needs Improvement (2)	Inadequate (1)
41	Ability to lay out the ideas of the presentation in an organized and clear flow that is appropriate to the technical audience				
42	Ability to lay out the ideas of the presentation in an organized and clear flow that is appropriate to non- technical audience (Motivation and Introduction parts).				
43	Demonstrate excellent understanding of the presented material.				
44	Illustrate well important concepts using the right examples and figures.				
45	Use effective body language and appropriate eye-contact; and listen to, respond to, and respect the audience				

Individual Assessment (Per Student) – Project Advisor – 15%

	Element	Outstanding (4)	Meets Expectations (3)	Needs Improvement (2)	Inadequate (1)
Teamwork (10%)					
46	Attend group meetings regularly and arrives on time, and contributes meaningfully to group discussions				
47	Complete group assignments on time and in quality manner.				
48	Demonstrate a cooperative and supportive attitude and contributes significantly to the success of the project.				
49	Share leadership, authority, power, and influence with team members.				
50	Plan and schedule tasks considering their priorities and dependencies (Work Breakdown Structure, WBS).				
51	Use project management best practices and tools (Define clear project roles, change management process, progress monitoring and reporting, quality assurance, ...)				
Presentation and Writing (5%)					
52	Ability to lay out the ideas of the presentation in an organized and clear flow that is appropriate to the technical audience.				
53	Ability to lay out the ideas of the presentation in an organized and clear flow that is appropriate to non- technical audience (Motivation and Introduction parts).				
54	Communicate the ideas in an organized flow in clear and readable style that is appropriate to non- technical audience.				
55	Demonstrate excellent understanding of the presented material.				
56	Use effective body language and appropriate eye-contact; and listen to, respond to, and respect the audience.				